Which Steam tag pair should an indie studio build? Horror × Multiplayer

MADT 7202 Mid-Term · Panjapol Ampornratana 6810424009 · Repo: github.com/hhej/steam-tag-whitespace

Steam is the largest PC game store. Players vote tags onto each game; the top tags drive store search.

1 The decision

An indie studio's creative director picks the two-tag concept for the next game. With no pair passing the rule in section 5, they would take a proven, crowded pair, e.g. Management × Simulation.

2 Why a network

Counts can't tell whether two tags attract different players. The network can: tags sharing few neighbours draw on audiences that rarely meet, so a game combining them can reach both.

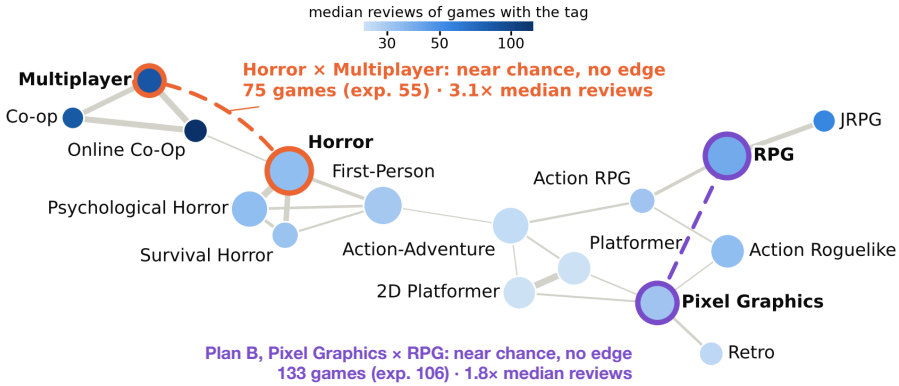
3 Data

Source	S1 = FronkonGames dataset (Hugging Face); S2 = live Steam store + reviews.
Obtained	S1 download (Sep 2025 snapshot). S2 scraped 2026-10-14: random 1,000 (seed 7202).
Legal	S1 MIT licence. S2: robots.txt allows /app/, /appreviews/; 1 req/1.5 s. No personal data.
Period	Released Jan 2023 – Dec 2024.
Volume	S1: all 9,094 games with ≥10 reviews. S2: 996.
Check	S1 vs S2: top-5 tags Jaccard 0.87; reviews ρ 0.98.
Bias	Favours games that sold; reviews count buyers, not unmet demand; late-2024 games had less time.

4 Graph construction

Node	A Steam tag. Undirected, weighted tag–tag projection of the game–tag bipartite graph.
Edge	Co-occurrence in a game's top 5 tags; suggests relatedness, not a stated relation.
Resolution	None: Steam tags come from a fixed list.
Weight	NPMI: co-occurrence vs chance, −1 to 1 (0 = chance). By raw count, 9 of the top-10 edges touch the 5 commonest tags; by NPMI, 0.
Window	Top 5, as the store page shows. With all 20, the pair's reviews vs the median game fall from 3.1× to 2.5× (200 games, mostly minor tags).
Excluded	"Indie" (seed); release-model, adult tags; tags on >25% of games: Singleplayer, Adventure, Action, Casual, 2D, 3D, Atmospheric, Exploration.
Pruning	Flat cutoffs: tag on ≥1.5% of games (136); edge NPMI ≥ 0.10 and ≥0.5% shared (45); pair ≥0.8% (73).
Size	439 tags → 81 nodes, 177 edges.

5 The picture



Horror's and Multiplayer's neighbours touch only via Online Co-Op. Dashed = no edge; node size = games.

Selection rule: among pairs on ≥73 games, keep those **not crowded** (games ≤ 1.5× chance, where chance = games with A × games with B ÷ 9,094) whose tags sit in **separate neighbourhoods** (≤20% shared neighbours); pick the one whose median reviews are highest relative to the median game (40).

Role	Pair	Games (chance)	Crowding	Shared neighbours	Median reviews
Pick	**Horror × Multiplayer**	75 (55)	1.4×	8%	124 (3.1× 40)
Plan B	**Pixel Graphics × RPG**	133 (106)	1.3×	0%	70 (1.8× 40)
Crowded	**Management × Simulation**	121 (32)	3.8×	8%	127 (3.2× 40)

6 Findings

- **Horror × Multiplayer** is used about as often as chance (NPMI 0.07, no edge), so the niche isn't crowded, yet its median game gets 3.1× the typical game's reviews ($p < 0.001$).
- 9 of the 10 highest-review pairs are crowded (>1.5× chance), e.g. Management × Simulation (3.8×).
- **Robust:** #1 at all 8 edge cutoffs; beats the typical game in 100/100 resamples and on S2. **Weak spot:** #1 in only 36/100 resamples, as it sits right at the 73-game minimum.

7 Recommended action

- Week 0: pitch Horror × Multiplayer at greenlight; plan B is Pixel Graphics × RPG.
- Weeks 1–6: prototype. Week 7: Coming Soon page with Horror, Multiplayer as first tags.
- Before full production: re-run on 2025 releases; plan B if over 1.5× crowded or out of the top 3.

8 What would change my mind

At 1.37× chance the pair is near the 1.5× crowding limit (crossed in 17/100 resamples). If 2025 releases cross it, I switch to plan B.